USN. | | | | | | | | | | [24MTSCSDE22]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY-2025

DEEP LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	**Marks**
1.	**a.**	How Deep Learning differs with Machine Learning? Explain your answer with suitable example.	CO 1	L2	**10**
	b.	What is single layer perceptron and how it differs with multi-layer perceptron?	CO 1	L2	**10**
2.	**a.**	Write short notes on the below terms with example: **i.** Perceptron **ii.** Learning Rate **iii.** Epoch **iv.** Batch Size **v.** Pre Activation Functions	CO 2	L3	**10**
	b.	"Adding Hidden Layer may affect the accuracy of your model", What is your opinion about the statement? You are supposed to write with an example to support your opinion.	CO 2	L3	**10**
3.	**a.**	What is object/image classification? How it differs with object detection?	CO 3	L2	**10**
	b.	Explain the concept of pooling (max pooling and average pooling) in CNNs. How does data flattening connect pooling layers to fully connected layers in CNNs?	CO 3	L2	**10**
		OR			
4.	**a.**	Where is convolution used in neural network? How is neuroscience helping CNNs perform better?	CO 3	L2	**10**
	b.	What is overfitting? How it can be handled in multi-layer perceptron? Explain about the dropout method in brief.	CO 3	L2	**10**
5.	**a.**	Describe the role of Regional CNN in object detection. What is its basic principle of working?	CO 4	L3	**10**

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

	b.	Explain the architecture of a CNN and the role of each component, including convolution layers, pooling layers, fully connected layers, and softmax.	CO 4	L3	**10**
		OR			
6.	**a.**	What is transfer learning? Draw the architecture of VGGNet with proper label and explain it.	CO 4	L3	**10**
	b.	Write shorts notes on: **i.** FCNN **ii.** MaskCNN **iii.** FasterCNN	CO 3	L3	**10**
7.	**a.**	What challenges may occur during the deployment of AI models? Discuss in brief.	CO 5	L2	**10**
	b.	Explain RCNN architecture in detail and list its applications.	CO 5	L2	**10**
		OR			
8.	**a.**	Write in brief about at least 5 applications of Deep Learning in real world.	CO 3	L2	**10**
	b.	Explain different Variants of the Basic Convolution Function with suitable examples.	CO 3	L2	**10**